

AlphaAgent

Quantitative Factor & Parameter Inventory

May 19, 2026

9

Active Agents

223

Total Factors

67

Static Params

156

Dynamic Params

Agent Summary

#	Agent	Factors	Static	Dynamic	Key Theories
1	Technical	40	22	18	Price Action, TDA, HMM, Options Greeks
2	Fundamental	42	28	14	Fama-French 5F, Piotroski, Altman, DCF
3	Sentiment	24	1	23	NLP/LLM, Behavioural Finance, Options Flow
4	Macro	36	0	36	FRED Macro, Yield Curve, Credit Cycle, Liquidity
5	Risk	21	8	13	EVT, GARCH, Hawkes, Monte Carlo, Kelly
6	Currency	12	6	6	FX Carry, PPP, EM Risk, Petro-currencies
7	Geopolitical	22	2	20	Risk Premium, Safe-Haven, Commodity Shock
8	Insider	17	0	17	EDGAR 4/13F, Kyle Lambda, Dark Pool, Congress
9	Volatility	9	0	9	GARCH, Stochastic Vol, SKEW, VVIX
TOTAL		223	67	156	30% static 70% dynamic

Agent 1 - Technical Agent

40 factors | 22 static | 18 dynamic | Price Action * TDA * HMM * Opt

Covers 40 factors across classical technical analysis, options theory, topological data analysis, Hidden Markov regimes, calendar seasonality, and cross-sectional momentum. 22 parameters are static (fixed academic formulas); 18 are dynamically computed from market data.

Factor / Indicator	Theory	Parameters	Type
RSI (14)	Oscillator	window=14, thresholds 30/70 equity 25/75 crypto	DYNAMIC
MACD (12/26/9)	Signal Crossover	fast=12, slow=26, signal=9	STATIC
Bollinger Bands (20/2)	Vol Compression	window=20, std=2.0, bw_narrow=20, bw_wide=40	DYNAMIC
SMA 50/200 + Golden/Death Cross	Trend Following	periods=50, 200	STATIC
EMA 9/21 Crossover	Momentum	periods=9, 21	STATIC
ADX (14)	Trend Strength	window=14, thresholds 20/40	STATIC
ATR (14)	Volatility Unit	window=14	STATIC
OBV + 20-SMA	Vol Confirmation	threshold=+/-2%	STATIC
VWAP	Fair Value	intraday level	STATIC
Stochastic %K/%D	Oscillator	window=14, smooth=3, thresholds 20/80	STATIC
52-Week Range Position	Momentum Context	<40% / <75% scoring	STATIC
ATR-Normalised Momentum	Risk-Adj Momentum	20-day, score=50+mom*15	STATIC
Bollinger Bandwidth Squeeze	Vol Regime	bw from settings (20/40)	DYNAMIC
Momentum Acceleration (3M delta)	Jerk Factor	3M now vs prior, score=50+accel*1.2	STATIC
Variance Risk Premium	Options Theory	ATM straddle IV vs 30d RV	STATIC
Hurst Exponent	Fractal Markets	>0.55=trend / <0.45=mean-revert	STATIC
Jegadeesh-Titman 12M-1M	Cross-Sectional	threshold>5% /<-5%	STATIC
HMM Market Regime (3-state)	Hidden Markov	Bull / Bear / Crisis states	DYNAMIC
Seasonality Factors	Calendar Anomaly	month-based scoring	DYNAMIC
TDA Persistent Homology H0/H1	Topological DA	Betti-1, H0 entropy	DYNAMIC
IV Skew (25-delta put-call)	Options	score=60-skew*2, clamped [10,90]	STATIC
Gamma Exposure (GEX)	Options Greeks	T=21/365, Rf=4.5%	DYNAMIC
Max Pain Level	Options Theory	min(call+put intrinsic) across strikes	DYNAMIC
Implied Correlation SPY/Stock	Dispersion	>0.8 limited alpha / <0.4 idiosyncratic	STATIC
Fibonacci Retracement	Market Geometry	23.6/38.2/50/61.8/78.6% levels	STATIC
Pivot Points (daily + weekly)	Price Structure	PP=(H+L+C)/3, R1/R2, S1/S2	STATIC
Multi-Timeframe Confluence	Signal Alignment	daily vs weekly RSI+MACD, max 4 signals	STATIC
Order Flow Imbalance (CLV)	Microstructure	5d avg, 10d/20d money-flow ratio	STATIC
Cross-Sectional Momentum Rank	Factor Investing	22d lookback, peers=5 from settings	DYNAMIC
Signal Backtest Calibration	Walk-Forward	lookback=126d, min_similar=5	DYNAMIC
Double Top/Bottom Patterns	Chart Patterns	60-bar scan, +/-3% tolerance	DYNAMIC
Volume Profile / POC	Market Profile	60-bar, 20 bins, 70% value area	DYNAMIC
PCA Signal Quality	Factor Analysis	consensus + dispersion across factors	DYNAMIC
Ichimoku Cloud	Japanese TA	Tenkan=9, Kijun=26, Senkou=52	STATIC
Chaikin Money Flow (14)	Vol/Price Flow	MFV 14-period	STATIC
TRIX (15)	Triple EMA Osc	period=15, %change	STATIC
Parabolic SAR	Trend Reversal	AF_step=0.02, AF_max=0.20	STATIC

Supply/Demand Zones	Order Flow	60-bar, vol spike >1.5x avg	DYNAMIC
Sector ETF Momentum	Relative Strength	5-day, 1.5x weighting	DYNAMIC
5-Day Price Momentum	Short-Term Mom	6-day return	STATIC

Agent 2 - Fundamental Agent

42 factors | 28 static | 14 dynamic | Fama-French 5F * Piotroski * Altman

Covers 42 factors across valuation multiples, accounting quality (Piotroski F-Score, Beneish M-Score, Altman Z-Score), growth metrics, Fama-French factor exposures, CAPM alpha, and LLM earnings-call analysis. 28 parameters use fixed academic thresholds; 14 are derived from live market/fundamental data.

Factor / Indicator	Theory	Parameters	Type
Piotroski F-Score (9-point)	Quality Accounting	9 binary checks; <=3=Poor 4-6=Fair 7-9=Strong	STATIC
Altman Z-Score	Bankruptcy Prediction	weights 1.2/1.4/3.3/0.6/1.0; <1.81 distress	STATIC
Beneish M-Score	Manipulation Detect	8-variable; >-1.78=likely manipulator	STATIC
P/E Ratio (trailing)	Value	pe_cheap=15, pe_fair=25, pe_exp=40 (rate-adj)	DYNAMIC
Forward P/E	Value	same thresholds as trailing	DYNAMIC
P/B Ratio	Value	pb_under=1.5, pb_fair=3.0	STATIC
PEG Ratio	Growth-Value	peg_under=1.0, peg_fair=2.0	STATIC
EV/EBITDA	Enterprise Value	cheap=10, fair=20, exp=30	STATIC
P/S Ratio	Revenue Multiple	cheap=2.0, fair=4.0	STATIC
FCF Yield	Cash Generation	good=2%, great=5%	STATIC
DCF Implied Upside	Intrinsic Value	upside_strong=30%, moderate=10%	STATIC
Revenue Growth YoY	Growth	strong=20%, moderate=10%	STATIC
Earnings Growth YoY	Growth	strong=20%, moderate=10%	STATIC
ROE	Profitability	strong=20%, moderate=10%	STATIC
ROA	Efficiency	strong=15%, moderate=8%, weak=3%	STATIC
Operating Margin	Quality	strong=25%, moderate=15%, weak=5%	STATIC
Current Ratio	Liquidity	liquid=2.0, adequate=1.5, tight=1.0	STATIC
Interest Coverage	Solvency	strong=10x, adequate=5x, weak=2x	STATIC
Asset Turnover	Efficiency	good=1.0, weak=0.5	STATIC
EPS Surprise	Earnings Quality	beat=+10%, miss=-2%	STATIC
Accruals Ratio	Earnings Quality	(NI-OCF)/Assets; <-0.05 / 0.05 / 0.15	STATIC
Net Margin Trend	Profitability Momentum	current vs prior net margin delta	STATIC
Dividend Cut Probability	Income Risk	payout>90% or (>70% and FCF<0)=cut risk	STATIC
Shares Buyback Signal	Capital Allocation	<-3% / <0% / +2% shares change	STATIC
Fama-French HML (Value)	Factor Investing	PE+PB+EV blend	STATIC
Fama-French SMB (Size)	Factor Investing	market cap deciles	STATIC
Fama-French RMW (Quality)	Factor Investing	ROE+margins+debt ratio	STATIC
Fama-French BAB (Low Vol)	Factor Investing	beta regime scoring	STATIC
CAPM Jensen's Alpha (1Y)	Asset Pricing	a=Ri-[Rf+B(Rm-Rf)], score=50+alpha*1.5	DYNAMIC
Dividend Growth CAGR (5Y)	Income	>10% / >5% / >0% thresholds	STATIC
Lockup Expiration Proxy	Supply Overhang	0-180d=25 / 180-365d=45 / >365d=65	STATIC
P/E vs Sector Median	Relative Value	<0.8=85 / <1.0=65 / <1.3=45	DYNAMIC
P/E vs 5Y Historical Avg	Mean Reversion	<0.85=80 / <1.0=60 / <1.3=40	DYNAMIC
Earnings Consistency (CoV)	Quality	CoV<0.2 / <0.4 / <0.7	DYNAMIC
CEO/CFO Departure (8-K NLP)	Event-Driven	90-day EDGAR window	DYNAMIC
M&A Activity (8-K NLP)	Event-Driven	180-day EDGAR window; >2=75 / >0=65	DYNAMIC

Earnings Revision Momentum	Estimate Trend	>5% / >2% / >-2% (30d/60d)	DYNAMIC
Graham Number	Benjamin Graham	$\sqrt{22.5 * EPS * BVPS}$; upside vs price	DYNAMIC
Earnings Call NLP (Gemini)	LLM Analysis	quality+revenue+guidance+red-flags (4 dims)	DYNAMIC
Net Debt / EBITDA	Leverage	<1.0=75 / <2.5=60 / <4.0=40	DYNAMIC
FCF Quality (FCF/NI)	Cash Conversion	>1.1=88 / >0.8=68 / >0.5=45	DYNAMIC
Analyst Revision Momentum	Estimate Sentiment	bull% change >8%=80 / >2%=65	DYNAMIC

Agent 3 - Sentiment Agent

24 factors | 1 static | 23 dynamic | NLP/LLM * Behavioural Finance *

Covers 24 factors including Gemini RAG news sentiment, Reddit/social analysis, options flow (unusual activity, put/call skew), AAll contrarian survey, Fear & Greed, short-squeeze scoring, Transfer Entropy, and exponential news-decay modelling. 23 of 24 factors are fully dynamic.

Factor / Indicator	Theory	Parameters	Type
News Sentiment (Gemini RAG)	NLP/LLM	0-100 scale (0=panic 100=euphoria)	DYNAMIC
Short Interest % + DTC	Behavioural	short_interest_high=20%, DTC threshold	DYNAMIC
Fear & Greed Index (contrarian)	Behavioural Finance	<=25 bullish / >=75 bearish	DYNAMIC
Analyst Consensus %	Sell-Side Sentiment	>70%=+5% / <30%=-4% probability	DYNAMIC
Market Breadth (RSP/SPY)	Breadth Analysis	>65=+3% / <35=-3%	DYNAMIC
Options Put/Call OI Ratio	Options Flow	<0.7=70 / 0.7-1.2=50 / >1.2=25	DYNAMIC
News/Social Momentum (3d)	Social Listening	base=40, mult=4.0, score=min(80,40+count*4)	DYNAMIC
EPS Forward Revision	Estimate Trend	>15%=85 / >5%=60 / >-5%=50	DYNAMIC
Unusual Options Activity	Smart Money	call vol/OI>2.0; score=min(85,40+ratio*30)	DYNAMIC
Transfer Entropy (news->price)	Information Theory	lag-1 cross-corr; causal signal if >0.3	DYNAMIC
Signal Entropy (Shannon)	Information Theory	H=(1-entropy)*80+10	DYNAMIC
Analyst Price Target Upside	Sell-Side	>20%=85 / >5%=65 / >-5%=50	DYNAMIC
AAll Sentiment (contrarian)	Behavioural	bull-bear spread <-20=75 / >30=25	DYNAMIC
Consumer Credit (margin proxy)	Macro-Sentiment	3M % change >2%=60 / <-1%=40	DYNAMIC
Short Interest Change MoM	Behavioural	<-10%=75 (covering) / >+10%=20 (rising)	DYNAMIC
Sentiment Momentum (3d vel.)	Social Velocity	3d vs 3-6d delta; >2=70 / >0=55 / <-3=30	DYNAMIC
Source Credibility Weight	Media Quality	WSJ/BBG=1.0, SeekingAlpha=0.70 (static tiers)	STATIC
Headline/Body Alignment (LLM)	NLP	Gemini dual-pass tone divergence check	DYNAMIC
FinBERT-Style NLP (Gemini)	LLM	6-dim: sentiment/revenue/mgmt/risk/catalyst/conf	DYNAMIC
Reddit Sentiment (WSB+Stocks)	Social Media	>60% bullish=72 / 45-60%=55 / <30%=35	DYNAMIC
News Decay Model (exp)	Signal Decay	halflife=4.6 days; dynamic near earnings	DYNAMIC
Unusual Options Sweeps	Smart Money	>3 calls: +4% prob / >3 puts: -4% prob	DYNAMIC
Short-Squeeze Score	Technical/Sentiment	short float% + days-to-cover; >20%+5d=80	DYNAMIC
PCA Signal Quality	Factor Analysis	factor consensus + dispersion	DYNAMIC

Agent 4 - Macro Agent

36 factors | 0 static | 36 dynamic | FRED Data * Yield Curve * Credit

All 36 factors are dynamically sourced - primarily FRED economic series (CPI, PCE, TIPS, PMI, claims, M2, retail sales, LEI) plus market proxies (VIX, credit spreads, copper/gold, global ETFs). The most dynamic agent - zero static parameters.

Factor / Indicator	Theory	Parameters	Type
Recession Probability	Macro Regime	FRED-derived; low=0.2 / cutoff=0.6	DYNAMIC
Yield Curve (10Y-2Y Spread)	Yield Curve	>0.5%=100 / inverted<0%=0; positive=0.5	DYNAMIC
Fed Funds Rate	Monetary Policy	easy=2.0% / tight=4.0%	DYNAMIC
CPI Inflation YoY	Inflation	target=2.5% / elevated=5.0%	DYNAMIC
PCE Inflation YoY	Inflation (Fed gauge)	target=2.0% / near_target=2.5% / elevated=3.5%	DYNAMIC
VIX (2Y rolling percentile)	Fear Index	percentile-based; fallback 90-(VIX-10)*2.2	DYNAMIC
Credit Spreads (HYG/LQD 1m)	Credit Cycle	score=50+delta*5	DYNAMIC
Copper/Gold Ratio (1m)	Growth Signal	>5%: +5% / <-5%: -5%	DYNAMIC
BTC Risk-On Signal (1m)	Risk Appetite	>20%: + / <-20%: -	DYNAMIC
Global Breadth (ACWI/SPY)	Global Macro	>2%: +3% / <-3%: -4%	DYNAMIC
Dollar Index Regime (DXY)	FX / Liquidity	STRONG=30 / WEAK=70	DYNAMIC
M2 Money Supply YoY	Liquidity	FRED M2SL; >5%=75 / >0%=50 / <0%=30	DYNAMIC
ISM PMI Manufacturing	Business Cycle	FRED NAPM; expansion=50 / strong=55 / weak=48	DYNAMIC
UMich Consumer Sentiment	Consumer Confidence	FRED UMCSENT; strong=90 / moderate=70 / weak=55	DYNAMIC
Initial Jobless Claims (4W avg)	Labour Market	FRED ICSA; low=220K / elevated=260K / high=350K	DYNAMIC
TIPS 10Y Breakeven Inflation	Inflation Expectation	FRED T10YIE; low=1.5% / high=2.5% / danger=3.0%	DYNAMIC
Amihud Illiquidity Ratio	Market Microstructure	stress_threshold=2.0x 3M avg	DYNAMIC
Bond-Equity Corr (TLT/SPY 60d)	Regime Detection	stress_corr=0.3	DYNAMIC
Semiconductor RS (SOXX/SPY 1m)	Sector Rotation	>5%: +5% / <-5%: -5%	DYNAMIC
Global Pre-Market (6 intl ETFs)	Global Macro	EWJ/EWZ/EWY/EWG/EWU/EWQ avg >3%	DYNAMIC
Fed Balance Sheet (WALCL 2M)	QE/QT Signal	>1%=75 (QE) / <-0.5%=30 (QT)	DYNAMIC
Growth vs Value Rotation	Factor Rotation	IWF/IWD 1m RS; >3%=growth / <-3%=value	DYNAMIC
Repo Stress (SOFR-FF spread)	Funding Markets	critical=50bps	DYNAMIC
Fed Rate Change Direction	Monetary Policy	3M delta >0.24pp=hiking / <-0.24pp=cutting	DYNAMIC
VIX 5-Day Change	Vol Momentum	score=50-delta*4	DYNAMIC
SPY vs SMA(200)	Market Regime	>5%=bull(80) / <-5%=bear(15)	DYNAMIC
Large vs Small Cap (SPY/IWM)	Cycle Phase	1m RS; >4%=late cycle / <-4%=early cycle	DYNAMIC
Sector RS (sector ETF/SPY)	Sector Rotation	1m RS	DYNAMIC
Contagion Corr Spike	Systemic Risk	SPY/TLT/GLD/HYG 20d avg corr vs historical	DYNAMIC
Real Interest Rate (10Y-TIPS)	Real Returns	>1.5%=70 / >0.5%=55 / >0%=45 / >-1%=30	DYNAMIC
Retail Sales ex-Auto MoM	Consumer Spending	FRED RSXFS; >1%=75 / >0%=60 / >-1%=40	DYNAMIC
Leading Economic Index (LEI)	Forward-Looking	FRED USSLIND 3M change; >0.5=72	DYNAMIC
HY Credit Spread OAS	Credit Cycle	FRED BAMLH0A0HYM2; dynamic vs 5Y median	DYNAMIC
NFCI Financial Conditions	Liquidity	FRED NFCI; neutral=0 / tight>0 / easy<0	DYNAMIC
Equity Risk Premium	Valuation	(E/P%) - Real Rate; >4%=75 / >2%=60	DYNAMIC
PCA Signal Quality	Factor Analysis	macro factor consensus	DYNAMIC

Agent 5 - Risk Agent

21 factors | 8 static | 13 dynamic | EVT * GARCH * Hawkes * Monte Carlo

Covers 21 risk factors: GARCH(1,1) volatility regimes, Extreme Value Theory (GPD) tail risk, Monte Carlo + Quasi-MC VaR, Hawkes jump process, Kelly sizing, black-swan detection, and override circuits for geo-shocks and carry unwinds.

Factor / Indicator	Theory	Parameters	Type
GARCH Vol Regime (1,1)	GARCH Volatility	fitted a/B/omega; pct 25/75/95	DYNAMIC
EVT Tail Risk (99% VaR)	Extreme Value Theory	min_obs=100, threshold_pct=10%, block=21d	DYNAMIC
Monte Carlo 95% CI (5d)	Stochastic Sim	paths=5000, GBM	DYNAMIC
Kelly Criterion (half-Kelly)	Optimal Sizing	fraction=0.5, cap=25%	DYNAMIC
Black Swan Detection (>5-sigma)	Fat Tail	sigma=5.0 threshold	DYNAMIC
Flash Crash Detection	Circuit Breaker	ticker=8%, SPY=6.5% 1-day move	STATIC
WAR/GEO Shock Override	Regime Filter	VIX=35 gold=2% oil=5% mult=0.35	STATIC
Carry Unwind Override (JPY)	Carry Trade Risk	USDJPY=125, multiplier=0.50	STATIC
KL Divergence (return shift)	Distribution Change	20d vs 252d baseline; threshold=1.0	DYNAMIC
Hawkes Process (jump cascade)	Point Process	branching warn=0.8, critical=0.95	DYNAMIC
Quasi-Monte Carlo VaR (Sobol)	Low-Discrepancy	5d horizon, 95% CI	DYNAMIC
Liquidity Risk (vol ratio)	Market Micro	high=1.5x, low=0.7x vs 20d avg	STATIC
Drawdown from ATH	Drawdown Analysis	2Y high; warning=30%	STATIC
Tail Ratio (95th/5th pct)	Return Asymmetry	positive=1.2, negative=0.8	STATIC
Correlation Regime (vs SPY 60d)	Regime Detection	delta_threshold=0.1	STATIC
Rolling Sharpe (63d)	Risk-Adj Return	annualised	DYNAMIC
Rolling Sortino (63d)	Downside Risk	annualised	DYNAMIC
Vanna/Charm Exposure	Options 2nd-order	vanna_neg_threshold=0.5	DYNAMIC
Return Skewness + Excess Kurtosis	Distribution Shape	skew<-1=20; kurt>5=warning	DYNAMIC
Options GEX Wall	Gamma Exposure	>0=absorb vol / <0=amplify	DYNAMIC
Pairs Trading Z-score	Statistical Arb	log-ratio of correlated pair; Z*15 scoring	DYNAMIC

Agent 6 - Currency Agent

12 factors | 6 static | 6 dynamic | FX Carry * PPP * EM Risk * Petro-

Covers 12 FX-driven factors: DXY regime with sector-weighted FX exposure, major pairs (EUR/USD, USD/JPY, USD/CNY, GBP/USD), carry-trade attractiveness, petro-currency signals (CAD/AUD), and EM currency stress via EEM proxy.

Factor / Indicator	Theory	Parameters	Type
DXY Regime + Sector FX Exposure	FX Carry	sector weights 0.10-0.55 (static) DXY dynamic	DYNAMIC
EUR/USD Rate	PPP / FX	>1.05=65 / <0.95=35	STATIC
USD/JPY Carry Signal	Carry Trade	strong=140, weak=120	STATIC
USD/CNY (China Stress)	EM Risk	stress=7.3	STATIC
GBP/USD Rate	FX	>1.25=65	STATIC
EM Currency Stress	EM Risk	USD 1m gain vs EM basket; >3%/-2%	DYNAMIC
FX Translation Impact	Revenue Impact	-DXY_1m * sector_exp * 0.25	STATIC
Petro-Currency (CAD/AUD 1m)	Commodity FX	>2%: +3% / <-2%: -3%	DYNAMIC
Carry Trade Attractiveness	Rate Differential	Fed - ECB; >1%=70 / >0%=50	DYNAMIC
Real Interest Rate (10Y-TIPS)	Real Returns	>1%=65 / >0%=50 / >-1%=40	DYNAMIC
EM FX Pressure (EEM 1m)	EM Flow	momentum-based	DYNAMIC
DXY vs SMA(50)	Trend Direction	>1% above=strong dollar headwind	DYNAMIC

Agent 7 - Geopolitical Agent

22 factors | 2 static | 20 dynamic | Risk Premium * Safe-Haven Flow

Covers 22 geopolitical factors: commodity/safe-haven flows (oil, gold, copper), defense/EM stress, credit contagion, shipping demand, election cycle phase, GPR Index (FRED), and NLP keyword-scanning for active conflicts, sanctions, and tariff/regulatory risks.

Factor / Indicator	Theory	Parameters	Type
Oil Shock (XLE/SPY RS 1m)	Commodity Risk	>5%: -8% / <-5%: +3%	DYNAMIC
Gold Safe-Haven (GLD 1m)	Safe-Haven Flow	>5%: -10% / <-3%: +5%	DYNAMIC
EM Stress (EEM/SPY 1m)	EM Contagion	<-5%: -6% / >3%: +4%	DYNAMIC
Defense Sector RS (ITA/SPY 1m)	War Risk Premium	>5%: -7%	DYNAMIC
VIX Fear Index	Fear Gauge	cap=80; >30: -10% / >20: -5%	DYNAMIC
Copper/Gold Ratio (1m)	Growth vs Safety	>3%: +5% / <-3%: -5%	DYNAMIC
Global Breadth (ACWI/SPY 1m)	Global Risk	<-3%: -4% / >2%: +3%	DYNAMIC
USD Safe-Haven (DXY 1m)	Currency Risk	>2%: -6% / <-2%: +4%	DYNAMIC
Credit Stress (HYG/LQD 1m)	Credit Risk	<-2%: -7% / >1%: +4%	DYNAMIC
Sector Rotation Signal	Risk Rotation	XLK*1.5 - XLE - XLF*0.5	DYNAMIC
Shipping Index (BDRY 1m)	Trade Demand	<-20%=demand destruction	DYNAMIC
Commodity Shock (XLE+XLB 1m)	Inflation Risk	>8%: inflation / <-8%: demand cooling	DYNAMIC
EM Contagion Risk (EMB 1m)	Contagion	<-3%=EM stress spreading	DYNAMIC
Oil Direct (BZ=F 1m)	Commodity	>15%: -5% / <-15%: +3%	DYNAMIC
Election Cycle Phase	Political Risk	2026 midterms / 2028 presidential (hardcoded)	STATIC
Transport Index (IYT/SPY 1m)	Dow Theory	Dow Theory transportation confirmation	DYNAMIC
Commodity Index (PDBC 1m)	Inflation	>12%: -4% prob	DYNAMIC
GPR Geopolitical Risk (FRED)	Geopolitical Risk	>2x hist: -8% / >1.5x: -4% / <0.75x: +3%	DYNAMIC
Active Conflict Score (NLP)	Text Mining	10 keywords; >3 hits: -6%	DYNAMIC
Sanctions Risk (NLP)	Text Mining	8 keywords; >=2 hits: -5%	DYNAMIC
Tariff/Regulatory Risk (NLP)	Text Mining	combined >3 keywords: -4%	DYNAMIC
PCA Signal Quality	Factor Analysis	geopolitical factor consensus	DYNAMIC

Agent 8 - Insider Agent

17 factors | 0 static | 17 dynamic | EDGAR 4/13F/13D * Kyle Lambda

All 17 factors are dynamic: EDGAR Form 4 cluster buys, 13F institutional change QoQ, activist 13D/G filings, Kyle's Lambda (price impact / microstructure), Congressional trading signal, FINRA dark pool print ratio, short-squeeze scoring, and ETF flow impact.

Factor / Indicator	Theory	Parameters	Type
EDGAR Form 4 Activity (60d)	Information Asymmetry	cluster threshold by market cap	DYNAMIC
Material Events (8-K, 30d)	Event-Driven	>=5=warning (30)	DYNAMIC
Insider Cluster Buys	Smart Money	score=min(85, 40+count*12)	DYNAMIC
13F Institutional Change QoQ	Institutional Flow	>2pp=75 (accum) / <-2pp=20 (distrib)	DYNAMIC
Short Squeeze Potential	Short Float	short%+days-to-cover combined	DYNAMIC
Float Reduction (buyback)	Capital Allocation	<-3% / <0% / <2% shares YoY	DYNAMIC
Kyle's Lambda (price impact)	Microstructure	normalized >2=thin(10) / <0.5=liquid(70)	DYNAMIC
Congressional Trading Signal	Political Intelligence	House Stock Watcher 180d; net>0=75	DYNAMIC
FINRA Short Volume %	RegSHO Data	weekly; >55%=25 / <45%=70	DYNAMIC
ETF Flow Impact	Passive Flow	sector ETF price*vol proxy	DYNAMIC
Activist 13D/G Filing (90d)	Activist Catalyst	EDGAR; >0=80 (activist signal)	DYNAMIC
Top-10 Holder Concentration	Ownership Structure	HHI; 0.05-0.25=60 / >0.25=40	DYNAMIC
Dark Pool Print Ratio (FINRA)	Institutional Flow	off-exchange%; >50%=72 / >35%=55	DYNAMIC
Insider Cluster (30d Form 4)	Form 4 Analysis	>=threshold=75	DYNAMIC
EDGAR Institutional Ownership	Ownership	% from EDGAR 13F data	DYNAMIC
Net Insider Shares (buy/sell)	Sentiment	net shares bought/sold	DYNAMIC
Analyst Price Target	Sell-Side	mean target vs current price	DYNAMIC

Agent 9 - Volatility Agent

9 factors | 0 static | 9 dynamic | GARCH * Heston * Stochastic Vol *

All 9 volatility factors are dynamic: GARCH regime, VRP (IV vs RV), put/call ratio, Kalman dynamic beta, 60d SPY correlation, VVIX (vol-of-vol), VIX term structure (contango/backwardation), realised vol trend ratio, and SKEW index.

Factor / Indicator	Theory	Parameters	Type
GARCH Vol Regime (1,1)	GARCH	fitted alpha/beta/omega; pct 25/75/95	DYNAMIC
Put/Call OI Ratio	Options Theory	bearish=0.7, overbought=1.2	DYNAMIC
IV vs Realized Vol (VRP)	Vol Risk Premium	underpriced=0.7, overpriced=1.5	DYNAMIC
Kalman Dynamic Beta	State-Space Model	Kalman filter vs SPY returns	DYNAMIC
SPY Correlation (60d rolling)	Systematic Risk	rolling 60d window	DYNAMIC
VVIX (vol of vol)	Second-Order Vol	extreme=120, elevated=90	DYNAMIC
VIX Term Structure	Vol Surface	VIX3M - spot VIX; >1=contango / <-1=backwardation	DYNAMIC
Realized Vol Trend (10d/30d)	Vol Momentum	calm=0.8, spiking=1.2 ratio	DYNAMIC
SKEW Index (tail risk)	Tail Risk Premium	extreme=140, elevated=130	DYNAMIC

Quant Engine Modules (23 modules)

Shared infrastructure powering all agents. Each module is a standalone quantitative library callable by any agent or the orchestrator.

Module (quant_engine/)	Theory / Model	Key Parameters
bayesian.py	Bayesian Updating	prior=0.5, sensitivity=1.8, corr_penalty=(1-corr)
garch.py	GARCH(1,1)	fitted a/B/omega per ticker; 1d/5d/10d forecasts
technical.py	10 Core Indicators	composite 0-100, bullish/bearish count
evt.py	Extreme Value / GPD	min_obs=100, threshold_pct=10%, block=21d
hawkes.py	Hawkes Point Process	branching warn=0.8, critical=0.95
scoring.py	Piotroski/Altman/Beneish	hardcoded academic formulas
calibration.py	Walk-Forward Calibration	lookback=126d, rsi_window=15
tda_signal.py	Topological Data Analysis	Persistent Homology H0/H1, Betti numbers
hmm.py	Hidden Markov Model	3-state: Bull/Bear/Crisis
kalman.py	Kalman Filter	dynamic beta estimation
kelly.py	Kelly Criterion	half-Kelly, vol-adjusted position size
monte_carlo.py	GBM Monte Carlo	paths=5000, dt=1/252
quasi_mc.py	Sobol Sequences (QMC)	low-discrepancy sampling
copula.py	Gaussian/Clayton Copula	tail dependence modelling
granger.py	Granger Causality	F-test, maxlag=10
causal_engine.py	Causal Inference	transfer entropy
heston.py	Heston Stochastic Vol	k/theta/sigma/rho fitted
sabr.py	SABR Vol Surface	alpha/beta/rho/nu fitted
rough_vol.py	Rough Heston	Hurst H<0.5 fractional Brownian
multifractal.py	MF-DFA	q-order multifractal moments
quantum_finance.py	Quantum Amplitude Est.	portfolio optimization
lob.py	Limit Order Book	microstructure metrics
factor_exposure.py	Risk Factor Loadings	Fama-French 5-factor exposures

settings.yaml - All Configurable Parameters (50 entries)

All numeric thresholds, windows, and multipliers defined in config/settings.yaml. These are loaded at startup and override hardcoded defaults. All are STATIC (set once at deployment) unless overridden via the /api/v1/settings endpoint.

Parameter Key	Value	Purpose	Type
technical_period	1y	Data lookback window	STATIC
risk_period	5d	Forecast horizon	STATIC
backtest_period	3y	Backtest window	STATIC
rsi_window	14	RSI calculation window	STATIC
bollinger_window / std	20/2.0	Bollinger window and multiplier	STATIC
adx_window / atr_window	14	ADX & ATR window	STATIC
stoch_window / smooth	14/3	Stochastic window & smoothing	STATIC
bw_narrow / bw_wide	20/40	Bollinger bandwidth thresholds	STATIC
num_paths	5000	Monte Carlo simulation paths	STATIC
ci_level_1/2/3	.68/.95/.99	Confidence interval levels	STATIC
regime_low/normal/high_pct	25/75/95	GARCH percentile regime boundaries	STATIC
put_call_bearish/overbought	0.7/1.2	Put-call ratio thresholds	STATIC
iv_rv_underpriced/overpriced	0.7/1.5	IV vs RV thresholds	STATIC
vix_elevated/extreme	20/30	VIX level thresholds	STATIC
vvix_extreme/elevated	120/90	VVIX thresholds	STATIC
skew_extreme/elevated	140/130	SKEW index thresholds	STATIC
branching_warn/critical	0.8/0.95	Hawkes branching ratio thresholds	STATIC
half_kelly_fraction	0.5	Kelly fraction applied	STATIC
max_kelly_cap	0.25	Maximum Kelly position cap	STATIC
altman_z_quality	3.0	Altman Z threshold for quality	STATIC
pe_cheap/fair/expensive	15/25/40	P/E valuation thresholds	STATIC
peg_undervalued/fair	1.0/2.0	PEG ratio thresholds	STATIC
roe_strong/moderate	20%/10%	ROE quality thresholds	STATIC
quality_value_blend	0.65	65% quality 35% value in probability	STATIC
yield_curve_positive	0.50%	Yield curve threshold (positive)	STATIC
yield_curve_inverted	-0.50%	Yield curve threshold (inverted)	STATIC
fed_rate_easy/tight	2%/4%	Fed rate regime thresholds	STATIC
recession_prob_cutoff	0.6	Recession probability trigger	STATIC
cpi_target/elevated	2.5%/5%	CPI thresholds	STATIC
pce_target/elevated	2%/3.5%	PCE thresholds	STATIC
pmi_expansion/strong/weak	50/55/48	ISM PMI levels	STATIC
claims_low/high	220K/350K	Initial claims thresholds	STATIC
sofr_critical_spread	50bps	Repo stress threshold	STATIC
bond_equity_stress_corr	0.3	Bond-equity stress correlation	STATIC
amihud_stress_threshold	2.0	Illiquidity stress multiple	STATIC
black_swan_sigma	5.0	Black swan detection sigma	STATIC
flash_crash_ticker_pct	8%	Flash crash single stock threshold	STATIC
geo_shock_vix/gold/oil	35/2%/5%	Geo shock trigger thresholds	STATIC
geo_shock_multiplier	0.35	Position size cap during geo shock	STATIC
carry_unwind_usdjpy	125.0	JPY carry unwind trigger	STATIC

usdjpy_strong/weak	140/120	JPY regime thresholds	STATIC
usdcny_stress	7.3	CNY stress threshold	STATIC
neutral_probability	0.5	Bayesian prior (starting point)	STATIC
long_threshold	0.53	Probability to trigger LONG signal	STATIC
short_threshold	0.47	Probability to trigger SHORT signal	STATIC
bayesian_sensitivity	1.8	Signal amplification factor	STATIC
news_decay_halflife_days	4.6	News sentiment half-life	STATIC
transaction_cost_bps	5	Simulated transaction cost	STATIC
short_interest_high	20%	High short interest threshold	STATIC
risk_free_rate	5%	Risk-free rate for calculations	STATIC

Grand Total Summary

#	Agent	Factors	Static	Dynamic	Key Quant Theories
1	Technical	40	22	18	Price Action, TDA, HMM, Options Greeks, Fibonacci
2	Fundamental	42	28	14	Fama-French 5F, Piotroski F-Score, Altman Z, CAPM, DCF
3	Sentiment	24	1	23	NLP/LLM, Behavioural Finance, Options Flow, Reddit
4	Macro	36	0	36	FRED Macro, Yield Curve, Credit Cycle, Global Liquidity
5	Risk	21	8	13	EVT, GARCH, Hawkes Process, Monte Carlo, Kelly Criterion
6	Currency	12	6	6	FX Carry, PPP, EM Risk, Petro-currencies
7	Geopolitical	22	2	20	Risk Premium, Safe-Haven Flows, Commodity Shock, NLP
8	Insider	17	0	17	EDGAR, Kyle Lambda, Dark Pool, Congressional Trading
9	Volatility	9	0	9	GARCH, Stochastic Vol, Heston, SKEW, VVIX
TOTAL		223	67	156	30% static 70% dynamic 23 quant modules

Static vs Dynamic Parameter Classification

STATIC (67 params / 30%)

Fixed academic formulas and configurable thresholds. Examples: RSI window=14, MACD(12/26/9), Fibonacci levels (23.6/38.2/50/61.8/78.6%), Piotroski/Altman weights, Bollinger std=2.0, Parabolic SAR AF=0.02, Kelly cap=25%, Flash crash trigger=8%. Changed only via settings.yaml or /api/v1/settings.

DYNAMIC (156 params / 70%)

Computed fresh for every signal from live data sources: FRED macro series, EDGAR filings (Form 4/13F/8-K/13D), yfinance OHLCV + options chains, GARCH-fitted parameters (alpha/beta/omega), LLM/Gemini outputs (news sentiment, earnings-call NLP), Reddit/social counts, congressional trading data, dark pool prints, options Greeks (vanna/charm/GEX), rolling correlations, distribution moments (skew/kurtosis), and all momentum/return computations.